

look “He is arrogant, but does not know enough to understand that he is not even wrong” (needless to say, Fisher was one of those who did not see the crisis coming). It was hard to get the message across that, even if econometric methods could track the effects of feedback loops in normal times (natural, since errors are small), such models said nothing about large disturbances. And I will repeat, large disturbances are everything in Extremistan.

The problem is that if I am right, Fisher’s textbook, and his colleagues’ textbooks, should be dispensed with. As should almost every prediction method that uses mathematical equations.

I tried to explain the problems of errors in monetary policy under nonlinearities: you keep adding money with no result ... until there is hyperinflation. Or nothing. Governments should not be given toys they do not understand.

- * The “a priori” I am using here differs from the philosophical “a priori” belief, in the sense that it is a theoretical starting point, not a belief that is nondefeasible by experience.
- * Interestingly, the famous paper by Reverend Bayes that led to what we call Bayesian inference did not give us “probability” but expectation (expected average). Statisticians had difficulties with the concept so extracted probability from payoff. Unfortunately, this reduction led to the reification of the concept of probability, its adherents forgetting that probability is not natural in real life.
- * The intelligent reader who gets the idea that rare events are not computable can skip the remaining parts of this section, which will be extremely technical. It is meant to prove a point to those who studied too much to be able to see things with clarity.
- * This is an extremely technical point (to skip). The problem of the unknown distribution resembles, in a way, Bertrand Russell’s central difficulty in logic with the “this sentence is true” issue—a sentence cannot contain its own truth predicate. We need to apply Tarski’s solution: for every language, a metalanguage will take care of predicates of true and false about that language. With probability, simply, a metaprobability assigns degrees of credence to every probability—or, more generally, a probability distribution needs to be subordinated to a metaprobability distribution giving, say, the probability of a probability distribution being the wrong one. But luckily I have been able to express this with the available mathematical tools. I have played with this metadistribution problem in the past, in my book *Dynamic Hedging* (1997). I started putting an error rate on the Gaussian (by having my true distribution draw from two or more Gaussians, each with different parameters) leading to nested distributions almost invariably producing some class of Extremistan. So, to me, the variance of the distribution is, epistemologically, a measure of lack of knowledge about the average; hence the variance of variance is, epistemologically, a measure of lack of knowledge about the lack of knowledge of the mean—and the variance of variance is analog to the fourth moment of the distribution, and its kurtosis, which makes such uncertainty easy to express mathematically. This shows that: fat tails = lack of knowledge about lack of knowledge.
- † A Gaussian distribution is parsimonious (with only two parameters to fit). But the problem of adding layers of possible jumps, each with a different probability, opens up endless possibilities of combinations of parameters.
- ‡ One of the most common (but useless) comments I hear is that some solutions can come from “robust statistics.” I wonder how using these techniques can create information where there is none.
- * One consequence of the absence of “typicality” for an event on causality is as follows: Say an event can cause a “war.” As we saw, such war will still be undefined, since it may kill three people or a billion. So even in situations where we can identify cause and effect, we will know little, since the effect will remain atypical. I had severe problems explaining this to historians (except for Niall Ferguson) and political scientists (except for Jon Elster). Please explain this point (very politely) to your professor of Near and Middle Eastern studies.